

Human-in-the-Loop Engineering

Designing Systems Where Agents Propose, Humans Approve

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Your guides today

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What this workshop is

This is not a talk you sit and watch. You sit **with us** while an AI coding agent builds a real service from a real PRD — and we stop the agent at every moment that matters.

Every stop is a decision we make as a room. Where should events go? What do we do about this contradiction? Should we accept this estimate? We don't move on until we've decided together.

The PRD is deliberately flawed — the same flaws you see every week. The agent is genuinely capable. The question is: **capable of what, decided by whom?**

What you walk away with

- A filled-in `CLAUDE.md` for a Java webhook-receiver project — yours to keep and adapt
- A partially-built, running receiver that demonstrates the architecture in action
- A working mental model for where HITL checkpoints belong in your own projects

Not a finished product. The point is the process.

The scenario

Maria K. is a Payments PM at a fictional company. She wrote a PRD asking engineering to fix a broken webhook integration with payment provider **PayHub**.

The PRD has every real flaw: silent assumptions, internal contradictions, deferred decisions, an optimistic timeline. It is a normal PRD.

We hand it to Claude Code and step in when human judgment is required — not to fix bad writing, but because some decisions belong to the room, not the agent.

Architecture — pre-decided

Path	Stack	Reason
Hot path	Pure Java 21, no framework	Sub-5s ACK, sustained throughput
Warm path	Spring Boot 3.x	Processing, retries, persistence — dev velocity dominates
Cold path	Spring Boot 3.x	Admin, reconciliation, DLQ

Left alone, Claude Code would default to Spring Boot everywhere. The first thing `CLAUDE.md` does is forbid that default on the hot path.

Agenda — 2 hours

Time	Block
0:00 – 0:05	Welcome
0:05 – 0:20	PRD walkthrough
0:20 – 0:55	Build <code>CLAUDE.md</code>
0:55 – 1:40	Drive Claude Code
1:40 – 1:55	Wrap-up
1:55 – 2:00	Buffer

We will **NOT** finish the implementation. We **WILL** finish the `CLAUDE.md`. That is the priority.

HITL beats — what we decide together

- Where to **store events**
- How to **map identities** between PayHub and the local system
- What to do about an **internal contradiction** in the PRD
- A **major architectural decision** the PRD doesn't mention
- Whether to accept the agent's **effort estimate**

Not staged. They emerge from the PRD.

The Cast

8 specialized agents

Each refusing to silently decide

Why a cast, not a single agent?

A general-purpose agent silently makes decisions that should never be made silently: which database, what SLA, whether the feature is ready to ship, what counts as tested. Those decisions belong to humans — or at minimum, in front of humans before anything is committed.

Splitting work across role-shaped agents doesn't make the system safer by making it slower. It makes it more honest by giving each agent explicit boundaries, escalation rules, and a documented refusal to override the humans who own the decision.

*The agent doesn't get less capable. The **system** gets more honest.*

The roster

product-owner

Stewards the PRD. Surfaces silent assumptions. Won't let engineering quietly resolve open product questions.

project-manager

Owens scope, schedule, risk. Pushes back on unrealistic timelines. Runs HITL checkpoints.

solution-architect

Architecture, ADRs, trade-offs. Refuses to make architecturally significant decisions silently.

backend-developer

JVM-first implementation. Java 21, Spring Boot 3.x, virtual threads, persistence, observability.

qa-lead

Test strategy, quality gates, release criteria. Won't declare quality "verified" without evidence.

qa-engineer

Hands-on test design and automation across functional, integration, perf, security, accessibility.

code-reviewer

Pre-merge correctness, security, OWASP, rule enforcement. Last reasonable check before merge.

devops-engineer

CI/CD, GitOps, Kubernetes, IaC, observability. Owns the production runtime story.

What they all share

- **Documented refusal points** — things each agent won't decide alone
- **Explicit escalation paths** — to the architect, the PO, the room
- **Project rules are binding** — `CLAUDE.md` wins; agents flag conflicts, never override
- **Severity discipline** — Critical · Major · Minor · Suggestion, never conflated
- **Evidence over taste** — every finding cites file, line, rationale

How HITL discipline survives contact with velocity pressure.

The Pipeline

PRD → epic → stories → done

A human gate on every arrow

The flow

Step	Command	Writes	Gate
1	<code>/prd:review</code>	analysis	classify open Qs: blocking / deferrable
2	<code>/prd:to-epic</code>	epic	architecture decisions (data store, ordering)
3	<code>/epic:decompose</code>	stories	identity mapping, assumed ACs
4	<code>/epic:estimate</code>	estimate	accept / cut scope / move date

All artifacts land in `work/` as plain markdown. Diffable in a PR. Take-home after the workshop.

The four-step macro

Every gate, every time:

1. **DETECT** — name the boundary. *"We're about to choose the data store."*
2. **PROPOSE** — 2–3 concrete options with trade-offs. Never one option dressed as a question.
3. **DECIDE — BLOCK** — `AskUserQuestion`. The command halts. No *"I'll assume X for now."*
4. **RECORD** — ADR in `work/decisions/NNNN-*.md`. Linked from the artifact's `hitl:` list.

*The **command** owns the BLOCK. Agents are advisory — they cannot put a question to the room, and they cannot write the artifact and skip the stop.*

Two loops, one protocol

Loop 1 — Decomposition

Things that block *planning*.

`/prd:to-epic` · `/epic:decompose`

→ data store, retention period, identity mapping.

Loop 2 — Implementation

Things visible *only in the code*.

`/story:start`

→ "this slice assumes X — true?" Invisible until the developer opens the story.

A story's `hitl:` list growing *after* `/story:start` is the visible proof loop 2 fired.

The Toolbelt

Making the agent context-efficient, deterministic, and persistent

Deterministic repo context

cxpak

Spends CPU cycles so you don't spend tokens. The LLM gets a briefing packet instead of a flashlight in a dark room.

→ Deterministic, reviewable repo context.

github.com/Barnett-Studios/cxpak

codeindex

Local-first codebase index engine. Builds a semantic index on your machine — no code leaves the box.

→ Retrieval layer for grounded answers on large repos.

github.com/askcodebase/codeindex

Token optimization — rtk

Rust Token Killer. A single Rust binary installed as a Claude Code `PreToolUse` hook.

Transparently compresses noisy Bash output **before** it reaches the model:

`cargo test` · `git status` · `find` · `grep` · `npm` · `docker` · `kubectl`

*`cargo test` with 262 passing tests: **4,823 → 11 tokens.***

*Average savings across 2,900+ commands: **~89%.***

Why HITL cares: longer sessions, cheaper iterations, less context pollution between human review cycles.

github.com/rtk-ai/rtk

Session memory — two takes

`claude-mem` — **auto-capture**

Records the session, compresses with Agent SDK, auto-injects context on restart. Works across Claude Code, Codex, Gemini CLI, Windsurf, OpenCode, Copilot.

→ *"I don't want to re-prime every morning."*

github.com/thedotmack/claude-mem

`memory-palace` — **deliberate, MCP-based**

Explicit `remember` · `recall` · `forget` tools over a knowledge graph with semantic search.

→ *"The human chooses what the agent knows long-term."*

github.com/jeffpierce/memory-palace

Choosing memory strategy is itself HITL

You want...	Pick
Frictionless continuity	<code>claude-mem</code> — auto
Auditable, curated knowledge	<code>memory-palace</code> — manual + MCP
Neither — every session clean	The <code>CLAUDE.md</code> itself

The discipline is not picking the fanciest tool. It is being explicit about what the agent assumes between sessions.

What you need to follow along

- Java 21+ installed (`java --version` shows 21+)
- Maven 3.9+ or Gradle
- **Claude Code** (Anthropic's CLI) installed and signed in
- An Anthropic API key — we distribute one in the room
- This repo cloned locally

Or just watch. Repo is public — replay everything after.

github.com/cefothe/hitl-engineering-jprime-2026-workshop

After the workshop

The `CLAUDE.md` you walk away with is **yours** — and so is `.claude/`: pipeline, agents, rules, templates. Drop the repo into any project. **The heading contract is the API.**

The same three-step recipe makes any `CLAUDE.md` work:

1. Decide what the agent should **assume** — `## Architecture constraints`
2. Decide what it should **ask** — `## HITL checkpoints`
3. Decide what it should **refuse** — out-of-scope and forbidden patterns

The artifact is small. The discipline is the point.

If you remember nothing else

- **Stop the agent in code, not in prompts.** Commands own the BLOCK; tool grants enforce the boundary.
- **Headings are the contract.** The pipeline reads `## HITL checkpoints` and `## Architecture constraints` literally — drop it into any repo.
- **Every gate produces an ADR.** Decisions outlive the meeting. The audit trail IS the discipline.

Make the discipline structural. The rest takes care of itself.

Let's begin.

Open the PRD. Meet Maria.

Questions during, not just at the end. Interrupt us.

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